

KARAIKAL, India

WMO: 433460

Lat: 10.917N

Lon: 79.833E

Elev: 7

StdP: 101.24

Time Zone: 5.50 (IND)

Period: 99-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	21.9	22.6	18.0	12.9	26.3	18.7	13.5	26.4	8.3	26.5	7.5	26.7	2.1	320	0.410

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	7.9	37.2	26.6	36.4	26.6	35.6	26.6	29.6	33.6	29.2	33.3	28.7	32.8	3.6	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
28.6	25.1	32.1	28.1	24.3	31.9	27.7	23.7	31.6	97.9	33.8	95.8	33.4	93.6	33.0	33.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.3	6.4	5.8	DB	20.2	39.2	0.5	0.9	19.8	39.8	19.5	40.3	19.2	40.8	18.8	41.4	(4)
(5)			WB	18.5	31.2	0.6	1.3	18.0	32.1	17.6	32.9	17.3	33.7	16.8	34.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	29.0	25.9	26.9	28.6	30.5	31.6	31.5	31.2	30.3	29.8	28.6	26.9	26.2	(6)
(7)		DBStd	2.37	1.01	1.13	1.15	1.13	1.49	1.28	1.11	1.36	1.11	1.53	1.19	1.17	(7)
(8)		HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)
(10)		CDD10.0	6936	494	472	576	613	670	645	657	631	594	576	507	501	(10)
(11)		CDD18.3	3894	236	239	318	363	412	395	398	372	344	318	257	243	(11)
(12)		CDH23.3	47267	1972	2337	3824	5000	5746	5433	5384	4809	4341	3713	2583	2122	(12)
(13)		CDH26.7	21207	298	590	1524	2628	3297	3059	2931	2424	2031	1453	642	331	(13)
(14)	Wind	WSAvg	3.0	3.5	3.0	2.6	2.6	3.0	3.3	3.3	3.0	2.7	2.2	2.9	3.6	(14)
(15)	Precipitation	PrecAvg	1243	37	19	21	29	48	38	41	78	85	218	410	228	(15)
(16)		PrecMax	1770	199	91	186	226	188	154	126	242	203	737	844	458	(16)
(17)		PrecMin	624	0	0	0	0	0	0	11	9	1	0	18	41	(17)
(18)		PrecStd	290	42	28	44	49	49	46	30	54	49	153	207	157	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	30.2	31.7	33.9	36.5	38.7	38.2	37.5	37.0	36.0	34.4	31.8	30.2	(19)
(20)			MCWB	25.2	26.5	27.5	28.1	27.5	26.3	26.4	25.5	25.7	26.3	26.1	25.6	(20)
(21)		2%	DB	29.3	30.8	32.9	35.0	37.3	37.0	36.5	35.9	34.7	33.4	30.8	29.5	(21)
(22)			MCWB	24.6	25.9	27.1	28.4	27.5	26.1	25.9	25.7	26.0	26.4	26.2	25.4	(22)
(23)		5%	DB	28.8	30.1	32.2	34.1	36.2	36.1	35.7	34.9	33.9	32.6	30.0	29.0	(23)
(24)			MCWB	24.3	25.1	26.7	28.5	27.7	26.4	25.9	25.7	26.2	26.4	25.8	25.1	(24)
(25)		10%	DB	28.3	29.6	31.6	33.4	35.1	35.2	34.8	34.0	33.1	31.7	29.4	28.4	(25)
(26)			MCWB	24.0	24.7	26.4	28.2	27.8	26.5	25.9	25.8	26.4	26.4	25.6	24.8	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	26.4	27.6	28.9	30.4	30.5	29.7	29.3	29.0	29.1	28.2	27.7	26.9	(27)
(28)			MCD B	28.8	30.8	32.6	33.8	34.9	33.8	33.8	32.7	31.8	32.0	30.2	28.9	(28)
(29)		2%	WB	25.7	26.6	27.9	29.6	29.6	28.9	28.5	28.0	28.2	27.6	27.0	26.3	(29)
(30)			MCD B	28.2	29.9	31.7	33.3	34.0	33.5	33.3	32.2	31.6	31.2	29.6	28.4	(30)
(31)		5%	WB	25.2	26.0	27.3	29.1	29.1	28.3	27.9	27.5	27.7	27.2	26.6	25.9	(31)
(32)			MCD B	27.6	29.1	31.1	32.9	33.6	32.7	32.6	31.7	31.3	30.6	29.0	28.0	(32)
(33)		10%	WB	24.7	25.4	26.8	28.5	28.7	27.8	27.2	27.0	27.2	26.8	26.2	25.5	(33)
(34)			MCD B	27.3	28.6	30.6	32.4	33.0	32.1	31.8	31.3	30.9	30.1	28.5	27.6	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	5.1	5.6	6.2	6.4	7.9	8.4	8.2	8.0	7.3	5.8	4.6	4.4	(35)
(36)			MCDBR	5.2	5.8	6.7	7.1	9.5	9.5	9.2	9.0	8.3	7.1	5.4	4.8	(36)
(37)			MCWBR	2.7	2.7	2.9	2.9	3.5	3.6	3.6	3.4	3.4	2.9	2.5	2.6	(37)
(38)		5% WB	MCD BR	4.7	5.4	6.4	6.7	7.8	8.4	8.4	8.0	7.4	6.2	4.8	4.3	(38)
(39)			MCWBR	2.8	2.9	3.1	3.1	3.5	4.1	4.1	3.7	3.5	2.9	2.5	2.7	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.455	0.448	0.483	0.499	0.583	0.518	0.526	0.516	0.505	0.523	0.504	0.495	(40)	
(41)		taud	2.247	2.262	2.166	2.178	1.945	2.153	2.123	2.160	2.193	2.135	2.163	2.149	(41)	
(42)		Ebn at Noon	850	872	844	822	741	781	778	798	814	794	798	801	(42)	
(43)		Edh at Noon	138	140	156	153	190	152	157	154	150	157	149	149	(43)	
(44)	All-Sky Solar Radiation	RadAvg	5.02	5.92	6.40	6.33	6.02	5.73	5.57	5.68	5.66	4.82	4.11	4.21	(44)	
(45)		RadStd	0.26	0.25	0.30	0.30	0.38	0.34	0.30	0.28	0.34	0.31	0.44	0.35	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)		+0.28	N/A	N/A	+0.86	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page